

K_{sp} FRQ #1

1. Answer the following questions relating to the solubility of the chlorides of silver and lead.

(a) At 10°C, 8.9×10^{-5} g of AgCl(s) will dissolve in 100. mL of water.

(i) Write the equation for the dissociation of AgCl(s) in water.

(ii) Calculate the solubility, in mol L⁻¹, of AgCl(s) in water at 10°C.

(iii) Calculate the value of the solubility-product constant, K_{sp} , for AgCl(s) at 10°C.

(b) At 25°C, the value of K_{sp} for PbCl₂(s) is 1.6×10^{-5} and the value of K_{sp} for AgCl(s) is 1.8×10^{-10} .

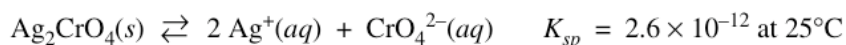
(i) If 60.0 mL of 0.0400 M NaCl(aq) is added to 60.0 mL of 0.0300 M Pb(NO₃)₂(aq), will a precipitate form? Assume that volumes are additive. Show calculations to support your answer.

(ii) Calculate the equilibrium value of [Pb²⁺(aq)] in 1.00 L of saturated PbCl₂ solution to which 0.250 mole of NaCl(s) has been added. Assume that no volume change occurs.

K_{sp} FRQ #2

1. Answer the following questions relating to the solubilities of two silver compounds, Ag₂CrO₄ and Ag₃PO₄.

Silver chromate dissociates in water according to the equation shown below.



- (a) Write the equilibrium-constant expression for the dissolving of Ag₂CrO₄(s).
- (b) Calculate the concentration, in mol L⁻¹, of Ag⁺(aq) in a saturated solution of Ag₂CrO₄ at 25°C.
- (c) Calculate the maximum mass, in grams, of Ag₂CrO₄ that can dissolve in 100. mL of water at 25°C.
- (d) A 0.100 mol sample of solid AgNO₃ is added to a 1.00 L saturated solution of Ag₂CrO₄. Assuming no volume change, does [CrO₄²⁻] increase, decrease, or remain the same? Justify your answer.

In a saturated solution of Ag₃PO₄ at 25°C, the concentration of Ag⁺(aq) is 5.3 × 10⁻⁵ M. The equilibrium-constant expression for the dissolving of Ag₃PO₄(s) in water is shown below.

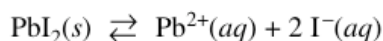
$$K_{sp} = [\text{Ag}^+]^3[\text{PO}_4^{3-}]$$

- (e) Write the balanced equation for the dissolving of Ag₃PO₄ in water.
- (f) Calculate the value of K_{sp} for Ag₃PO₄ at 25°C.
- (g) A 1.00 L sample of saturated Ag₃PO₄ solution is allowed to evaporate at 25°C to a final volume of 500. mL. What is [Ag⁺] in the solution? Justify your answer.

Ksp FRQ #3

1. Answer the following questions that relate to solubility of salts of lead and barium.

- (a) A saturated solution is prepared by adding excess $\text{PbI}_2(s)$ to distilled water to form 1.0 L of solution at 25°C . The concentration of $\text{Pb}^{2+}(aq)$ in the saturated solution is found to be $1.3 \times 10^{-3} M$. The chemical equation for the dissolution of $\text{PbI}_2(s)$ in water is shown below.



- (i) Write the equilibrium-constant expression for the equation.
 - (ii) Calculate the molar concentration of $\text{I}^{-}(aq)$ in the solution.
 - (iii) Calculate the value of the equilibrium constant, K_{sp} .
- (b) A saturated solution is prepared by adding $\text{PbI}_2(s)$ to distilled water to form 2.0 L of solution at 25°C . What are the molar concentrations of $\text{Pb}^{2+}(aq)$ and $\text{I}^{-}(aq)$ in the solution? Justify your answer.
- (c) Solid NaI is added to a saturated solution of PbI_2 at 25°C . Assuming that the volume of the solution does not change, does the molar concentration of $\text{Pb}^{2+}(aq)$ in the solution increase, decrease, or remain the same? Justify your answer.
- (d) The value of K_{sp} for the salt BaCrO_4 is 1.2×10^{-10} . When a 500. mL sample of $8.2 \times 10^{-6} M$ $\text{Ba}(\text{NO}_3)_2$ is added to 500. mL of $8.2 \times 10^{-6} M$ Na_2CrO_4 , no precipitate is observed.
- (i) Assuming that volumes are additive, calculate the molar concentrations of $\text{Ba}^{2+}(aq)$ and $\text{CrO}_4^{2-}(aq)$ in the 1.00 L of solution.
 - (ii) Use the molar concentrations of $\text{Ba}^{2+}(aq)$ ions and $\text{CrO}_4^{2-}(aq)$ ions as determined above to show why a precipitate does not form. You must include a calculation as part of your answer.